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Improving the architecture of the expert system for route optimisation and searching for data in GIS databases

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***Abstract*—The search for an optimal path from point A to point B in a GIS database can be carried out using various means and techniques. Here we propose an improvement to the architecture of the expert system to optimise this search. We are interested in the architecture and ordinary operation of an expert system in order to make improvements that will enable it to search for data more quickly by integrating the Breadth-first search algorithm and then optimally by using convolutional neural networks or CNNs to process images by extracting relevant information. It is capable of responding to a user's needs instantly and in real time.**

***Index Terms*—Big Data ; Map-reduce ; Classification ; SIG ; Expert system, Machine learning, Convolutional neural networks, Dijkstra algorithm.**

I. Introduction

The development of expert systems has never ceased. This branch of artificial intelligence offers great potential in numerous sectors such as banking, insurance and health.

In Burkina Faso, for example, the services offered by health centres face a number of challenges arising from the quality, lack or unavailability of certain services or equipment, and the limited number of places and/or staff. The search for the nearest site eligible for the availability of resources in a grid will enable a rapid response to a user request formulated from a device.

There are several approaches for determining the nearest site to a given object, to find the shortest path in a graph. The notions of 8-neighbourhood and 4-neighbourhood [1] , the use of regular grids by a quasi-affine transformation [2] and Dijkstra's famous algorithm [3] , to name but a few, are among the studies that enable us to find a site closer to an object, except that they do not do so in an expert system approach.

In this article, we undertake a study of these different algorithms in the context of an expert system. How we can set up an expert system that optimises routes and searches for data in a database.

The remainder of this article is organised as follows: in section II, we present the preliminaries that will enable us to better understand the subject, section III details the implementation of an expert system: the components as well as the methodology, in section IV we show our approach to improving expert systems, finally in section V we conclude our observations and discuss avenues for future research.

II. Preliminaries

To fully understand the subject and issues addressed in this work, it is necessary to define certain concepts. We'll choose a few of the most important to explain.

1) Expert system: In the field of artificial intelligence, a Knowledge Based Expert System (KBES) is a tool designed to model the reasoning of a human specialist, by reproducing his or her cognitive mechanisms. The software package performs reasoning based on known facts, and on rules implemented at the outset. It can then draw deductions or conclusions, and explain how the results were obtained (Jean-Charles Pomerol).

The first expert system was invented in 1965 by computer scientists Edward Feigenbaum and Bruce Buchanan, physician Joshua Lederberg and chemist Carl Djerassi. Called Dendral, it was capable of identifying the chemical constituents of a material from mass spectrometry and nuclear magnetic resonance results.

2) SIG: A geographic information system (GIS) is an information system designed to collect, store, process, analyze, manage and present all types of spatial and geographic data.

- information is stored clearly and definitively ;
- manage a multiplicity of attribute information on objects ;
- understand phenomena, forecast risks (simulations) ;
- create rapid maps ;
- locate in space and time ;

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- react rapidly to events that have an impact on the territory ;
- calculate costs and benefits ;
- involve a wider range of partners in planning choices ;
- provide adapted itineraries and plans.

3) Map-Reduce: MapReduce [4] is a simplified programming model for processing large quantities of data sets, developed by Google for data-intensive applications. The model is surprisingly simple and effectively supports parallelism [5]. Such a model is being adopted by Hadoop implementations, which are spreading rapidly and becoming a dominant force in the big data arena [6]. MapReduce enables an inexperienced programmer to develop parallel programs and create a program that can use computers in a cloud. In most cases, programmers only need to execute two functions, namely the map and reduce functions, which are commonly used in functional programming. The mapper considers the key/value pair as an input and generates intermediate key/value pairs, and the reducer merges all pairs associated with the same (intermediate) key and then generates an output. The map function is applied to each input (key1, value1), where the input domain is different from the generated output pair list (key2, value2). The elements of the list (key2, value2) are then grouped by a key. After grouping, the list (key2, value2) is divided into several lists [key2, list (value2)], and the reduction function is applied to each list [key2, list (value2)] to generate a final result list (key3, value3) [7].

4) Convolutional neural networks: A convolutional neural network (CNN) is a particularly effective neural network architecture for computer vision and image processing [8]. Convolutional neural networks are distinguished from other neural networks by their superior performance with image, speech or audio signal inputs [9]. They have three main types of layer:

- Convolution layer ;
- Pooling layer ;
- Fully Connected (FC) layer.

Convolutional neural networks power the tasks of image recognition and computer vision, which is a field of artificial intelligence (AI) that enables computers and systems to extract relevant information from images, videos and other digital visual inputs, in order to take action based on these inputs. The architecture of a convolutional neural network is as follows (Fig. 1.):

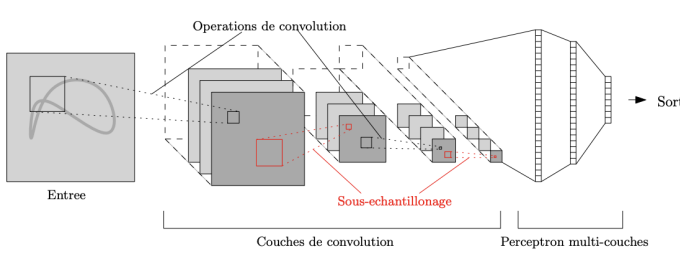


Fig. 1: CNN architecture

5) Dijkstra's algorithm: Dijkstra's algorithm [10] is used to solve the shortest path problem. This problem has several variants. The simplest is as follows: given a non-oriented graph, whose edges have weights, and two vertices of this graph, find a path between the two vertices in the graph, of minimum weight. Dijkstra's algorithm solves a more general problem: the graph can be oriented, and we can designate a single vertex, and ask for the list of shortest paths for all other nodes in the graph. The algorithm is named after its inventor, Dutch computer scientist Edsger Dijkstra, and was published in 1959.

III. Expert system architecture

The architecture of an expert system can vary according to the specific needs of the application domain. Some expert systems may include machine learning techniques to improve their performance over time. The implementation of an expert system generally involves the use of a reasoning algorithm to process the information contained in the knowledge base and generate conclusions or recommendations.

A. The components

The various components of an expert system are :

- Knowledge Base (KB):
  - Facts: Stores factual information about the domain of expertise ;
  - Rules: Contain conditional rules in the form of "IF condition THEN action" that express expert reasoning in the domain.
- Inference engine :
  - Reasoning system: Uses the rules of the knowledge base to deduce new information or conclusions from available facts ;
  - Forward chaining: Starts with known facts and derives conclusions by applying the rules ;
  - Backward chaining: Starts with a goal and works backwards to determine the necessary facts.

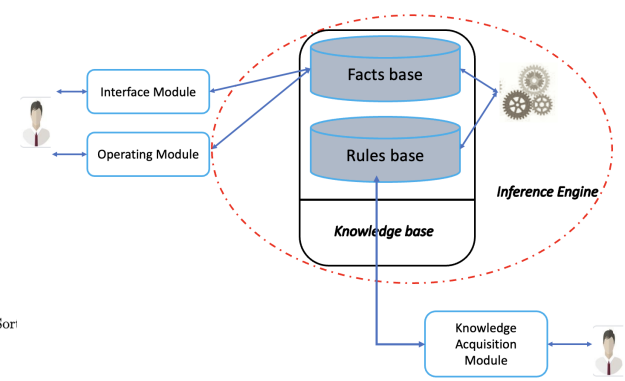


Fig. 2: Expert system architecture [11]

- User interface :
  - Consultation interface: Allows the user to ask questions, enter data and obtain advice or solutions ;

  - Graphical or natural language interface: Facilitates interaction with the system for non-technical users.
- Fact management engine :
  - Manages the fact base: keeps track of current facts and updates them as new information becomes available.
- Rules management engine :
  - Manages the rules base: Allows rules to be added, deleted or modified as required.
- Explanation module :
  - Provides explanations: Explains the reasoning followed by the system to reach a particular conclusion. This contributes to transparency and user understanding.
- Case base (optional) :
  - Store specific cases: For expert systems based on case retrieval, where solutions are based on similarities with previous cases.
- Knowledge acquisition mechanisms :
  - Facilitates the addition of new knowledge: Allows human experts to add, modify or delete rules and facts.
- Certification module (optional) :
  - Checks the validity of conclusions: Can be included to guarantee the reliability of results.

B. The steps

Generally speaking, when setting up an expert system, the following stages are observed :

- Initialisation :
  - Load the knowledge base (facts and rules) ;
  - Setting up the inference engine.
- User input :
  - Receive input from the user, usually in the form of initial facts or questions.
- Processing initial facts :
  - Add initial facts to the fact base.
- Reasoning (Inference Engine):
  - Use the inference engine to apply rules from the knowledge base ;
  - Perform forward or backward chaining to deduce new information ;
  - Update the fact base accordingly.
- Checking output criteria :
  - Check that output criteria (objectives) have been met.
- Answer generation :
  - If output criteria are met, generate response or recommendation ;
  - If not, request more information from the user to refine the reasoning.
- Repetition (iteration):
  - Repeat the process according to the new information provided by the user.
- Explanation (optional) :
  - If the system has an explanation module, generate explanations of how the system reached its conclusions.
- End of session :
  - End the expert system session.

There are a few additional points specific to certain expert systems :

- Rules management engine: The algorithm must support the addition, deletion or modification of rules as required. This may involve dynamic management of the rule base ;
- Case base: If the expert system uses a case base, the algorithm must include mechanisms for retrieving and adapting relevant cases ;
- Machine learning (optional): Some expert systems incorporate machine learning techniques to improve their performance over time. This may require an incremental learning process.

C. Use cases for expert systems in road optimization

In the field of route optimization and data retrieval in a GIS database, many works exist [12], [13], [14]. However, there is very little work on the optimization and retrieval of data from a GIS database using expert systems.

In [9], A. SERE et al. propose a framework for searching data in a NoSQL database based on GIS data, containing an abstract table that could be inherited or specialized in any adopted GIS solution, leading to central data management instead of installing multiple database sites. The central database accepts data updated in the back-office by the data owner and enables data retrieval based on meshing techniques and the map-reduce algorithm in the front-office. Various meshing techniques have been presented to group GIS data with associated map-reduce content definitions to improve processing time. In this work, the focus is on searching for data in a GIS database. New techniques to improve data retrieval in a large database are proposed.

In his doctoral thesis [15], A. Poulhès takes a different approach to the problem. He focuses on the explanation, comprehension and retrieval of information in a transport system, by modeling all the actors and elements of the system, in what is known as multi-agent modeling. As the data in this transport system is considered to be GIS data, we find ourselves in the situation of searching for data in a GIS database.

According to the study by fasterCapital [12], the idea is to collect the GIS data supplied by the various sensors, make them available and leave it up to the user to choose the strategy to be implemented to optimize his or her route on the basis of the GIS data collected progressively within a given perimeter. This strategy is known as RTTC (Real-Time Traffic Control).

This pragmatic approach doesn't offer the user a clear-cut solution.

IV. Methodology

Researchers have already looked at different ways of searching for data in GIS databases [9], [16], [12], [13] optimizing routes according to a number of criteria. However, few have made use of expert system techniques. It is at this level that we have decided to place our study. We are proposing an improved architecture for an expert system that will enable route optimization and data retrieval from GIS databases.
Our objectives are to :

- To study the concepts of appropriate distances and suitable grids for grouping data;
- Propose a structuring of the intelligent NOSQL database showing GIS data, taking into account notions of probability (following the work of Sere et al [9] and more generally machine learning or AI techniques to improve queries);
- Propose the content of map/reduce functions [14] taking into account notions of probability and machine learning or AI techniques.

All of this should make it possible to search for a place based on a number of criteria on a map, by being positioned at a given location and finding the most optimal path or route to get there. All this through automatic learning.
In the work by A. SERE et al. [21] entitled "Data Research in Smart GIS Database using Decision Tree and Bayesian Probability", the question of structuring the intelligent database and proposing the content of the mapreduce function has already been solved to our satisfaction. We now need to focus on the third point, which concerns the architecture of the expert system.
Antoine Crochet-dumais in 2022 explains the architecture of an expert system in the following image:

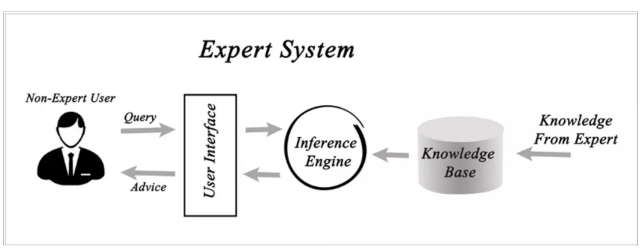


Fig. 3: Expert system architecture according to Antoine Crochet-dumais

This is where rule-based reasoning (rule base), fact evaluation, forward and backward chaining, uncertainty management (Bayesian networks), decision-making and explanation of conclusions take place.
In the light of what has already been done in the field of expert systems, route optimization and data retrieval in a GIS database, and given the complexity of the model, we feel

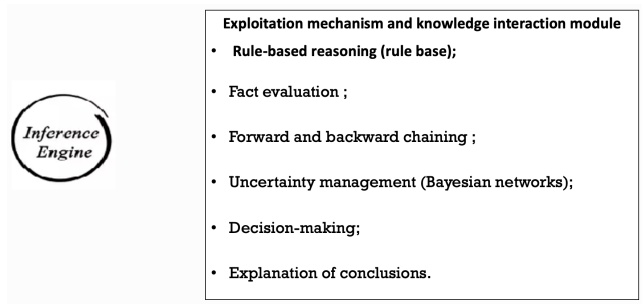


Fig. 4: The inference engine

that the use of computer vision at the level of the inference engine can enable us to achieve the objectives we have set ourselves. When we speak of computer vision, we necessarily mean neural networks.

Convolutional Neural Networks, or CNNs [17] for short, can be used to extract relevant information from images, videos and other digital visual inputs. Thus, by teaching them to detect roads on a map, to predict whether or not there is an obstruction on a road, and to detect what type of obstruction it is if any, they will enable expert systems to make optimal route proposals to join position A to position B.

The CNN model to be used is a *U-Net* [23] with a *resnet34* encoder pre-trained with a semantic segmentation architecture. The general structure of this architecture is as follows :

- **Encoder (contracting path) :** extracts features by progressively reducing the resolution (thanks to resnet34 here) ;
- **Bottleneck :** dense central layer where the image is highly compressed ;
- **Decoder (expanding path) :** reconstructs the mask by increasing the resolution, layer by layer ;
- **Skip connections :** between the encoder and decoder to help preserve detail.

Neural networks will be used to identify routes on a map image, and then the application of Dijkstra's [10] algorithm to determine the shortest path seems likely to produce satisfactory results. For in addition to the complexity of the model, there's also the fact that the variables are non-linear.

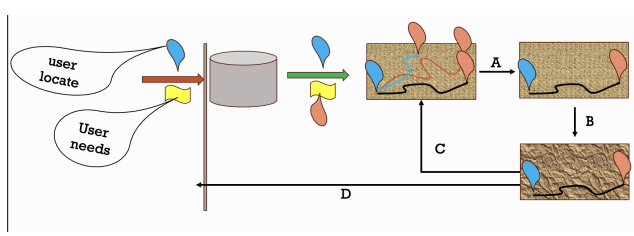


Fig. 5: Methodology in pictures

The example of health care provision is more appropriate in this context. Let's say a patient in a given location wants to find a health center offering a particular service. He will

query the database to find his need. His query contains as parameters his geographical position and his need. As a response to his query, he receives all the locations offering the service he requires, along with their geographical coordinates.

- A : We apply Dijkstra's algorithm to find the shortest path;
- B : With the geographic coordinates of the two (2) positions, we request a real-time satellite image of the area;
- C :Computer vision (CNN) is applied to the received image to extract routes and detect any obstructions. If an obstruction is detected, the section is eliminated and the shortest path is calculated again.
- D : Returns the response containing the site and the most optimal route for patient access.

Figure 6 shows the proposed expert system architecture.

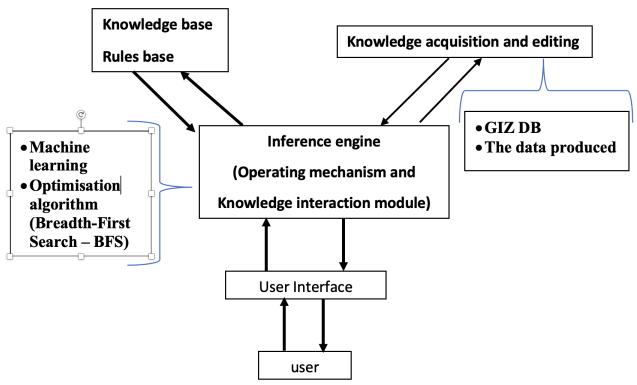


Fig. 6: Proposed expert system architecture

The knowledge acquisition and editing component will provide the data on which the system will learn. This data will essentially be GIS data and map images. It will come from sensors, cameras and other technologies to collect data on traffic conditions, such as vehicle speeds, volumes and travel times. In the knowledge base and rule base component, it will be no different from that of an ordinary expert system. It stores factual information on the domain of expertise, in this case the data produced by the various centers of interest. If we stay with the example of the health sector, this will be the data produced by the health centers.
The big difference with other expert systems, the innovation we bring, is in the inference engine. We're using machine learning and neural networks to improve the algorithm [16] for optimizing data search routes. The algorithm used here is the Breadth-first search algorithm, which applies Dijkstra's algorithm to obtain the shortest path.

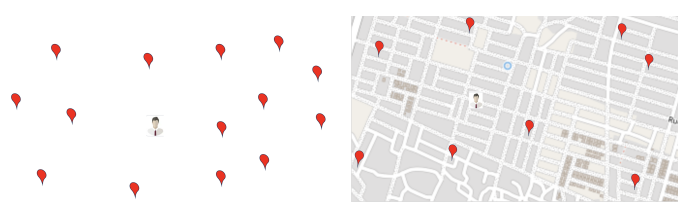
V. Implementation and Discussion

Like all expert systems, we start by initializing it. Ours is initialized by inserting the map of the area into the NoSQL database, in this case MongoDB. This map will list as many key sites as possible. This data will be in the form of a collection.

Algorithm 1 Card and point insertion

An example of a point :
Pharmacie = {
"type": "Feature",
"geometry": {"type": "Point", "coordinates": [40.7128, -74.0060] },
"properties": {"name": "Pharmacie"} }
An example of a polygon (a map) :
carte = {
"type": "Feature",
"geometry": { "type": "Polygon",
"coordinates": [[[40.7128, -74.0060], [40.7308, -73.9975], [40.7358, -73.9910], [40.7128, -74.0060]]] },
"properties": { "name": "Burkina Faso" } }

The knowledge base consists of the various points listed on the map, each with its coordinates and properties. The rules base is just the fact of being able to match each point to a user need.
Once the user's need has been sent, the aim of our system is to satisfy it by enabling the user to find the closest point to the point from which he is making his request, and the most optimal path to take to get there and find his need. Initially, the user will use his interface to indicate his need and his geographical position. This need is then transmitted to the knowledge base, which is responsible for finding all the structures that have this need, as shown in the first image (a) of figure 4. All these structures are geolocated and their coordinates entered into a graph. If we inject these structures into a map, we obtain something similar to the second image (b) in figure 4.



(a) Available locations (b) Map locations

Fig. 7: Geolocation of various locations inserted in a map

From this map, we extract an undirected graph with the user's position as vertex (Fig. 5.). All vertices that have the user's need are accessible from the vertex. The default weight of the stops in this graph is the distance between each vertex and the other.
In real life, with our implementation, we obtain figure 8, which gives a single location containing the requirement.

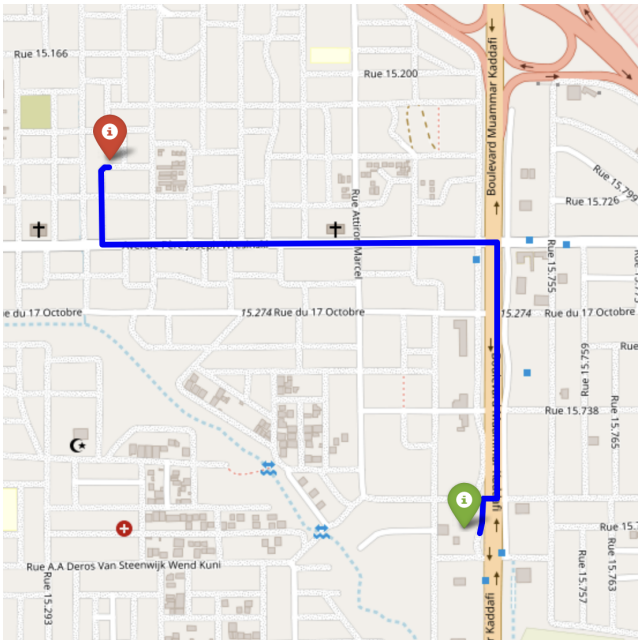


Fig. 8: Route to a requirement

From this point on, Dijkstra's algorithm is applied to determine the shortest path our user can take to reach his need. Applying this algorithm amounts to doing the same thing as googlemaps [22] ; which is not the end of our method. At this point, we call up satellite images of the same geographical area at the same time. With this image, we apply the convolutional neural network (CNN) algorithm, which focuses on roads to see if there is an obstruction at a given level. If, however, the road is blocked, then it considers this road to be impassable and recalculates the shortest path by eliminating this road. If no obstruction is found, this route is considered the most optimal.

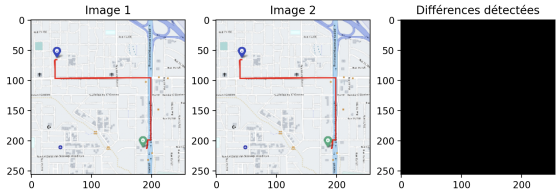


Fig. 9: Image extraction and comparison

In our example, we found no obstacles on this route. So we can inform the user that the path is optimal for finding his need.

In the course of implementing our methodology, we were faced with a number of challenges. One of these was obtaining satellite images in real time. Many platforms offer them (google maps, NASA Worldview, Here maps ...) but these images remain difficult to access or are accessed late. Then there's the quality of satellite images. The higher the quality, the easier and more accurate it is to detect obstacles. Another no less important difficulty is differentiating between different types of obstruction for specific treatments. Whether it's a traffic jam, an incident, a sudden gathering of people, water on the road, foliage on a tree, or any other obstacle, the treatment must not be the same. There's also the distance between the needs and the user's position (zoom reduction before image playback).

VI. Conclusion and perspective

The implementation of an expert system involves necessarily the use of a reasoning algorithm to process the information contained in the knowledge base and rules in order to generate conclusions or recommendations. The architecture of an expert system varies according to needs and field of application. Some include machine learning techniques to enrich their knowledge base and improve their performance in proposing conclusions and recommendations. Here, we propose an architecture that includes a sophisticated search algorithm to find the shortest path quickly and optimally, and a means of analyzing images and extracting important information to help draw conclusions.

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